

1. Administrative & Study Identification

Full Study Title: A Study to Learn About Medicine Called Ritlecitinib in Children Aged Between 6 to 12 Years With Severe Alopecia Areata **Official Title:** A PHASE 3 RANDOMIZED, DOUBLE-BLIND, PLACEBO-CONTROLLED STUDY TO INVESTIGATE THE EFFICACY AND SAFETY OF RITLECITINIB IN PEDIATRIC PARTICIPANTS 6 TO LESS THAN 12 YEARS OF AGE WITH SEVERE ALOPECIA AREATA **Short Title/Acronym:** Allegretto Study **Protocol Number:** B7981027 **NCT Number:** NCT07029711 **Posting ID:** 499424325523173823 **Trial Status:** Recruiting **Sponsor/Company:** Pfizer **Investigational Product:** Ritlecitinib (Trade Name: Litfulo) **Phase of Development:** Phase 3 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of ritlecitinib compared to placebo in pediatric participants (6 to <12 years of age) with severe alopecia areata (AA) on regrowth of lost scalp hair. **Secondary Objectives:** To evaluate the safety, tolerability, acceptability, and palatability of ritlecitinib, and to evaluate its effect on patient-centered outcomes.

2.2 Background & Rationale Alopecia areata (AA) is an autoimmune condition causing sudden-onset hair loss. Ritlecitinib (Litfulo), an oral, selective dual inhibitor of JAK3 and the TEC family kinases, is currently approved to treat severe AA in adults and adolescents aged ≥ 12 years. However, there are no approved treatments for children aged <12 years. This study aims to address this unmet medical need by evaluating the safety and efficacy of ritlecitinib in pediatric patients with severe AA.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1:1 allocation ratio (One-third Ritlecitinib higher dose, one-third Ritlecitinib lower dose, one-third Placebo)

3.2 Planned Enrollment & Duration Target N: At least 225 participants **Number of Sites:** Global (At least 30% of the total study population will be recruited from Europe) **Treatment Duration:** 6 months (24 weeks) **Estimated Study Start:** Q3 2025 **Estimated Primary Completion:** Q4 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 6 to <12 years with a diagnosed AA (including alopecia totalis [AT] and alopecia universalis [AU]). 2. At least 50% scalp hair loss due to AA (Severity of Alopecia Tool [SALT] score ≥ 50) at both screening and baseline visits, without evidence of terminal hair regrowth within the previous 12 months. 3. Documented evidence of having received varicella vaccination (2 doses), OR evidence of prior exposure to varicella zoster virus (VZV) based on serological testing (i.e., a positive VZV Immunoglobulin G [IgG] antibody result) at screening. 4. For study participants in the EU/UK only: History of clinical response failure to AA treatment (such as topical, off-label pharmacologic, or hairpiece prosthetics).

4.2 Exclusion Criteria 1. Other (non-AA) types of alopecia, including any known congenital cause of AA. 2. Pre-existing hearing loss. 3. Any present or history of malignancies or lymphoproliferative disorder such as Epstein-Barr virus (EBV) related lymphoproliferative disorder, history of lymphoma, history of leukemia, or signs and symptoms suggestive of current lymphatic or lymphoid disease. 4. Clinically significant depression per PROMIS Parent Proxy Short Form - Depressive symptoms (T-score ≥ 70). 5. Any evidence of untreated or inadequately treated active or latent *Mycobacterium tuberculosis* (TB) infection; history (one or more episodes) of severe or serious cytomegalovirus (CMV) infection, herpes zoster (shingles) or disseminated herpes simplex; infection with hepatitis B virus (HBV) or hepatitis C virus (HCV). 6. Vaccination with live attenuated replication-competent vaccine within 6 weeks of first dose of study intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ritlecitinib higher dose, Ritlecitinib lower dose, or matching Placebo. **Route of Administration:** Oral (Capsule taken once each day at home). **Duration of Treatment:** 24 weeks. Participants completing the 24-week treatment period may have the option to enter the long-term extension (LTE) Study B7981028.

5.2 Concomitant & Prohibited Therapy Permitted: Standard allowable concomitant therapies for underlying health. **Prohibited:** Vaccination with live attenuated replication-competent vaccines within 6 weeks of the first dose of study intervention.

6. Study Timeline & Visit Schedule

During the 6-month study duration, participants will have 8 study visits at the study clinic and receive about 8 phone calls from the study team.

Visit ID	Window (Days)	Key Activities/Procedures
1	Up to 5 weeks	Informed Consent/ Assent, Medical History, SALT assessment, Blood/Infectious disease tests (e.g., VZV IgG)
2	Day 0	Randomization, First Dose, Baseline Questionnaires
3	Up to Week 24	8 clinic visits and ~8 phone calls for ongoing evaluation of safety, tolerability, PK sparse sampling, and efficacy assessments
4	Week 24	Final Efficacy Assessment, Exit Interview, or Transition to LTE Study (B7981028)
5	4 to 5 weeks	4-week off-treatment follow-up period for non-LTE participants

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. For US and Countries Following US Analysis Plan: Response based on achieving an absolute SALT score ≤ 20 . Defined as the difference in proportions of participants with the SALT ≤ 20 response at Week 24 between each ritlecitinib dose and placebo groups. **2. For EU/UK and Countries Following EU/UK Analysis Plan:** Response based on achieving an absolute SALT score ≤ 10 . Defined as the difference in proportions of participants with the SALT ≤ 10 response at Week 24 between each ritlecitinib dose and placebo groups.

7.2 Secondary & Exploratory Endpoints 1. **Change from baseline (CFB) in SALT score** (For all countries). 2. **Response based on achieving absolute SALT score ≤ 20 at all visits**, except for that included as the primary endpoint (For all countries). 3. **Patient Global Impression of Change (PGI-C) response** defined as a score of "moderately improved" or "greatly improved" at Week 24 (For EU/UK and Countries Following EU/UK Analysis Plan). 4. **Additional efficacy assessments: Eyebrow and eyelash assessments.** 5. **Additional patient-reported outcomes: Alopecia Areata Patient Priority Outcomes (AAPPO), PROMIS Parent Proxy - Anxiety Short Form 8a and Depressive Symptoms Short Form 6a, Behavior Rating Inventory of Executive Function®, Second Edition (BRIEF®2), and modified Children's Dermatology Life Quality Index (CDLQI).** 6. **Safety, tolerability, acceptability, and palatability tracking** (Incidence of AEs, SAEs). 7. **Pharmacokinetics (PK)** of ritlecitinib evaluated using sparse sampling.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via periodic study visits, specifically tracking for known and potential risks of ritlecitinib (e.g., infections). **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to regularly evaluate pediatric safety signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Target enrollment of $N = 225$ is appropriately powered to detect a statistically significant difference in the proportion of participants achieving the target SALT response between each ritlecitinib dose and the placebo group. **Handling of Missing Data:** Standard imputation methods established for missing PROs and SALT scores.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence and safety data integrity. **Audit Trail:** eCRF locking, tracking of clinician SALT grading, and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: B7981027 **Registry Number:** NCT07029711 **Posting ID:** 499424325523173823 **Sponsor/Lead Organization:** Pfizer **Company:** Pfizer **Study Title:** Allegretto Study **Official Regulatory Title:** A PHASE 3 RANDOMIZED, DOUBLE-BLIND, PLACEBO-CONTROLLED STUDY TO INVESTIGATE THE EFFICACY AND SAFETY OF RITLECITINIB IN PEDIATRIC PARTICIPANTS 6 TO LESS THAN 12 YEARS OF AGE WITH SEVERE ALOPECIA AREATA **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/litfulo-epar-product-information_en.pdf

12. Clinical Framework (Alopecia Areata Specific)

Primary Condition: Severe Alopecia Areata (AA) **Preferred UMLS Name:** Alopecia Areata **Disease Areas:** Severe Alopecia Areata **MeSH Code:** D000523 **SNOMED ID:** 402513000 **Diagnosis Threshold:** SALT score ≥ 50 ($\geq 50\%$ scalp hair loss) **Medical Methodology:** Targeted Immunomodulation (Dual JAK3/TEC family kinase

inhibition) **Optimization Target:** Hair regrowth (SALT score ≤ 20 or ≤ 10)

13. Study Procedures & Methodology

Management Pathway: Targeted oral systemic therapy (Ritlecitinib)
Education Protocol (HCP): Standardization training on the administration of the Severity of Alopecia Tool (SALT) and other assessments
Education Protocol (Patient): Counseling on daily capsule ingestion, completion of PROs, and reporting of any signs of infections
Quality Audit Mechanism: Centralized data review for SALT assessments and laboratory tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to 33-34 weeks (5 weeks screening, 24 weeks treatment, 4 weeks follow-up for non-LTE)
Onsite Visit Cadence: 8 clinic visits over 6 months
Remote Monitoring Frequency: Approximately 8 phone calls from the study team
Checklist Review Interval: Scheduled at specific trial visits

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				